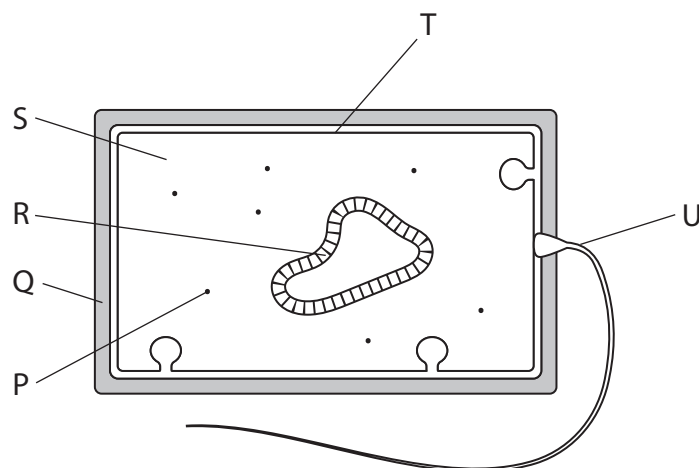


2 Some bacteria are photosynthetic.

Bacteria do not have chloroplasts.

(a) The diagram shows a photosynthetic bacterium.



(i) Which labelled structure contains the genes coding for the enzymes involved in photosynthesis?

(1)

- ☐ A P
- ☐ B Q
- ☐ C R
- ☐ D T

(ii) Where do the light-dependent reactions take place?

(1)

- ☐ A Q
- ☐ B S
- ☐ C T
- ☐ D U

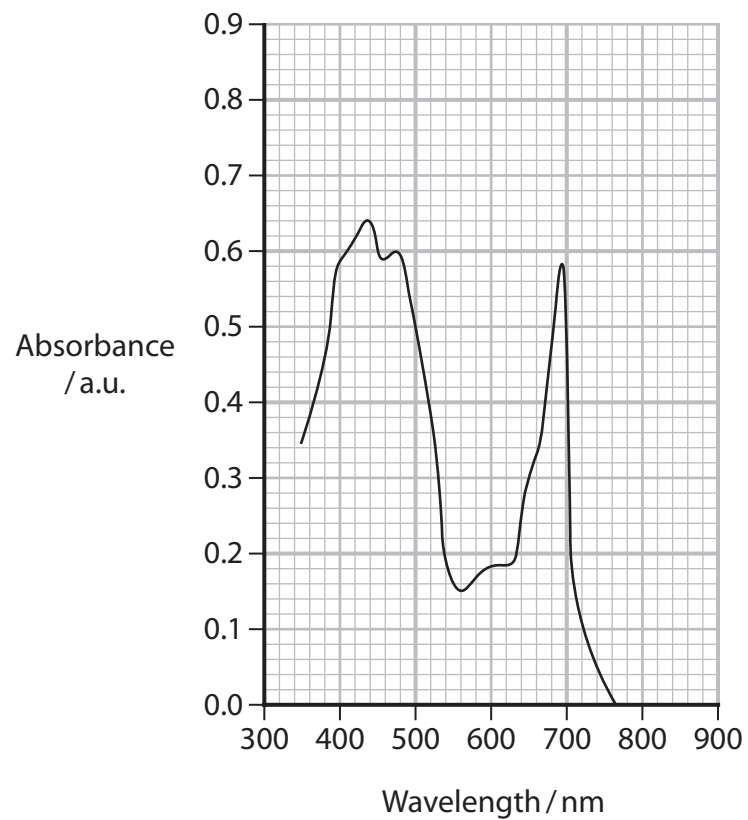
(iii) Where do the light-independent reactions take place?

(1)

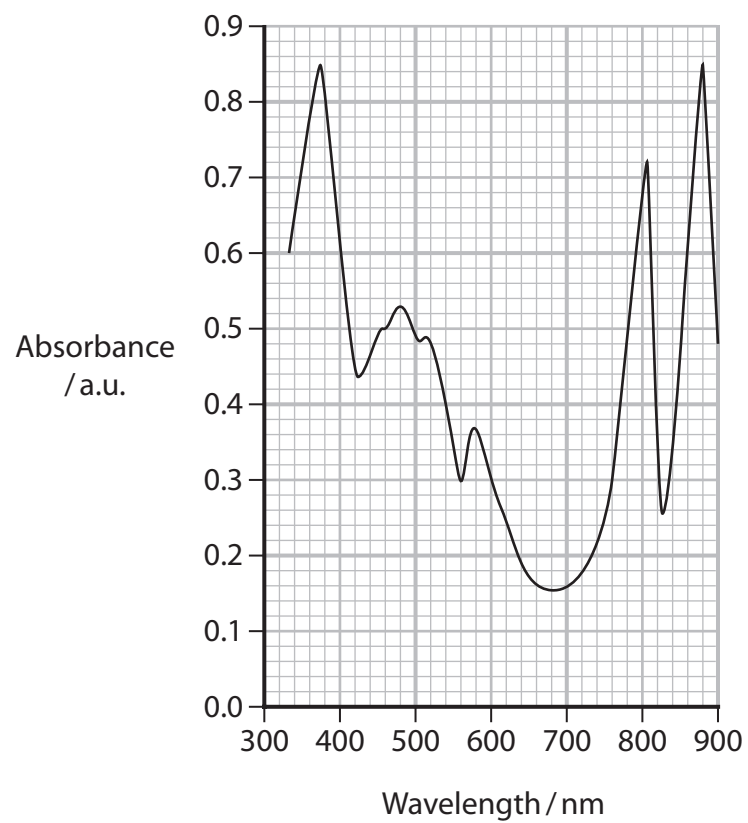
- ☐ A Q
- ☐ B S
- ☐ C T
- ☐ D U

- (b) The graphs show the absorption spectrum of the photosynthetic pigments in a plant and in a bacterium.

Plant



Bacterium



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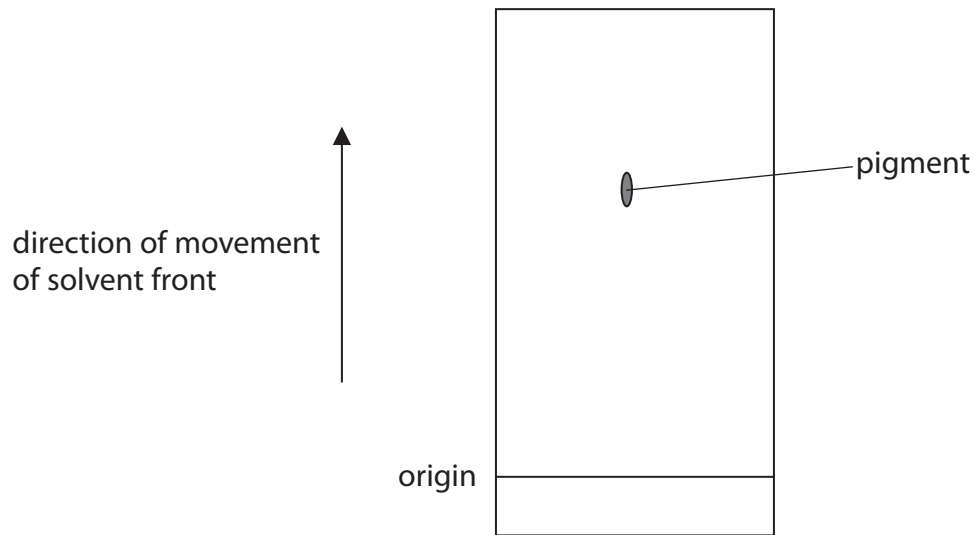
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Describe **two** conclusions that can be made about the differences in absorption spectra shown in these two graphs.

(2)

(c) Photosynthetic pigments can be identified using chromatography.

The diagram shows part of a chromatogram.



Measurements are taken from a chromatogram to determine the R_f value for a photosynthetic pigment.

Describe the measurements taken and how they can be used to calculate the R_f value for this pigment.

(3)

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(Total for Question 2 = 8 marks)

